Does Trade Openness Encourage Clean Energy Investment? Evidence from Emerging Markets

ECON F311 — International Economics

BITS Pilani, Pilani Campus

International public clean energy financing commitments, 2000–2023

Group 17

Ritik Yadav — 2024B3A30940

Sriansh Rajan — 2024B3A80901P

Nidhish Kumar Tyagi — 2024B3A80952P

Heet Jain — 2024B3A80929P

Jaivardhan Singh — 2024B3A40859P

12 October 2026

Abstract

Does a more open economy attract more clean energy investment? Trade can lower the cost of imported equipment and widen access to technology and finance, but it can also increase demand for fossil fuels. This report examines the question using annual data for fifteen selected emerging economies from 2000 to 2023. The outcome is international public financing commitments for clean energy, taken from UN Sustainable Development Goal indicator 7.a.1 and measured in constant 2023 US dollars per person. These commitments do not cover all domestic or private investment, and they are not the same as money actually disbursed. World Bank data supply trade openness, income, FDI, and population. Pooled, random effects, and country and year fixed effects models are compared, alongside diagnostic and sensitivity checks. The baseline fixed effects estimate is positive: a ten percentage point increase in the previous year's trade ratio is associated with a 0.108 increase in log one plus financing per person. The country-clustered p-value is 0.012, but the wild-cluster bootstrap gives 0.073. The estimate stays positive without China and in the pre-pandemic sample. It is not statistically significant when the sample starts in 2010, country trends are included, or first differences are used. A separate analysis also finds no clear link with solar and wind generation growth. The evidence points to a possible relationship, but it does not show that opening trade causes higher investment. Policy conclusions should therefore focus on specific equipment and financing channels rather than assume that a higher overall trade ratio will deliver more clean energy projects.

Keywords: trade openness; clean energy finance; emerging markets; panel data; renewable investment

1. Introduction

A renewable energy project can depend on several countries before it produces any electricity. Solar modules may be imported, turbine parts may come from foreign suppliers, and specialised engineering services may be sourced abroad. Trade can lower the cost of these inputs and give developers more choice. This is one reason to expect a link between openness and clean energy investment.

There are reasons the link may be weaker than it first appears. Trade can also support fossil fuel imports and energy-intensive manufacturing. Cheap equipment will not solve expensive credit, uncertain electricity contracts, or limited grid access. Public lenders may choose projects according to development needs or donor priorities. To examine the relationship, the financing measure must be clearly defined and lasting country differences must be separated from changes over time.

Antweiler et al. (2001) separate trade's environmental effects into scale, composition, and technique channels. Their evidence concerns pollution, not renewable finance, so its conclusion cannot be transferred directly. Copeland and Taylor (2004) situate trade and growth in the environmental literature. Applied here, these frameworks suggest that aggregate openness combines competing mechanisms and need not capture targeted liberalisation of renewable equipment.

This report brings together country-year financing data, tests its relationship with the previous year’s trade openness, and checks how the result changes under different assumptions. The data and calculations are retained so they can be examined. The main contribution is this comparison across selected countries and specifications, rather than a claim that the topic itself is new.

The available outcome is international public financing commitments for clean energy. It is not a measure of all money spent on investment. A commitment may be disbursed later, and domestic or purely private projects are largely outside the data. The analysis therefore answers a narrower version of the question in the title.

2. Literature Review

The environmental trade literature explains why the sign of the relationship is ambiguous. Antweiler et al. (2001) investigate pollution concentrations using scale, composition, and technique effects; their published abstract reports a net pollution-reducing result in their setting. This provides a mechanism framework rather than evidence about the financing measure used here. Copeland and Taylor (2004) offer a broader review of trade, growth, and environmental outcomes. Neither source warrants assuming that aggregate trade must raise renewable investment.

Eyraud et al. (2013) study green investment and its determinants. The related IMF working paper summarises historical evidence from advanced and emerging economies and identifies growth, financing conditions, fuel prices, and policy instruments as relevant factors (Eyraud et al., 2011). This motivates income and financial-context controls. The working paper and journal article belong to the same research line and are treated as related evidence rather than independent replications.

Polzin et al. (2015) examine public policy and renewable investment in an OECD panel. That setting differs from this emerging-market recipient-country analysis. It highlights the relevance of policy conditions, which country fixed effects can capture only when they are stable. Changing procurement rules, support schemes, and financing conditions remain possible omitted variables in the present design.

Ebaidalla (2024) directly connects taxation, technological innovation, trade openness, and renewable investment in its research question. Appiah-Otoo et al. (2025) examine governance, financial openness, and renewable investment in Africa. Financial openness is distinct from the goods-and-services trade ratio used here. These studies motivate attention to taxation, technological change, governance, and financing conditions. The comparison here concerns their research scope; detailed findings from unreviewed full texts are not attributed to them.

This study examines whether a within-country relationship appears in a public clean energy financing panel and whether it survives controls for trends, alternative periods, and inference with few countries. The outcome, selected sample, and reproducibility package distinguish this exercise from the cited studies. The contribution lies in the selected financing measure and sensitivity analysis, rather than a claim that the topic has not been studied.

3. Data and Methodology: Sources and Sample

The sample contains Brazil, China, India, Indonesia, Mexico, Türkiye, South Africa, Thailand, Malaysia, the Philippines, Viet Nam, Egypt, Morocco, Colombia, and Peru. It is a purposive selection across Latin America, Asia, and Africa, rather than an exhaustive emerging-market index list. Viet Nam is included for regional coverage. The results should not be generalised to every emerging or developing economy.

The analytical period is 2000–2023. World Bank data for 1999 provide the first preceding-year controls. The UN series also contains 2024 financing observations, but the main panel uses the period fixed during collection of the controls; those later observations are preserved in the raw responses and are not used. All 360 candidate observations are available for the main regression.

The dependent variable uses SDG indicator 7.a.1, series EG_IFF_RANDN. OECD and IRENA compile international public support for clean energy research and renewable energy production, including hybrid systems. The retrieved series records recipient-country commitments in millions of constant 2023 US dollars. Source footnotes identify donor-country deflators and exchange-rate conventions. No additional CPI conversion is applied.

The technology dimension ALL is selected once per country and year, with reporting type G. It is not added to its technology components, which would double count the flows. Each raw country response has 200 records: 25 years and eight technology categories. The main sample contains thirteen explicitly reported zero aggregate commitments. These zeros remain in the model; missing values are not converted into zeros.

The outcome covers public international finance, not total domestic and private investment, and measures commitments rather than actual expenditure. The current data source labels observations

Nature = E; the source status is retained without relabelling them as audited project transactions. UN methodology and source footnotes are archived. Duplicate keys, pagination completeness, units, base period, technology filters, and sample inclusion are checked in the scripts.

3.1 Variables, Sources, and Rationale

The trade ratio records actual imports and exports; it does not directly measure tariffs. Commodity prices, exchange rates, and changes in GDP can move the ratio even without a policy change. Values above 100% are possible because gross trade is compared with domestic value added. Dividing openness by ten makes its coefficient easier to read: it corresponds to a ten percentage point change.

Real income per person is measured in constant 2015 US dollars and enters in logarithms. Development can influence renewable demand, financing capacity, and eligibility for concessional support. FDI is a broad financing-context measure, not a measure of renewable-sector investment. Both controls are lagged by one year. Lagging reduces contemporaneous overlap but does not establish exogeneity.

Population converts international commitments into comparable per-person amounts. Dividing by GDP would combine the outcome with another denominator and would require compatible real-price definitions. The principal model therefore uses per-person commitments. A total-finance specification with a population control checks whether this scale choice drives the result.

Table 1. Variables and data sources

Variable	Definition / units	Source and rationale
Clean energy finance	ln(1 + commitments in 2023 US$/person)	UN SDG 7.a.1; principal outcome
Trade openness	Exports plus imports / GDP × 100; lagged	WDI NE.TRD.GNFS.ZS; exposure
Real income/person	Natural logarithm; lagged	WDI NY.GDP.PCAP.KD; development
FDI inflows	Net inflows / GDP × 100; lagged	WDI BX.KLT.DINV.WD.GD.ZS; finance context
Population	Persons	WDI SP.POP.TOTL; scale adjustment
Country/year effects	Country intercepts; annual indicators	Stable heterogeneity/common shocks

3.2 Data Transformations and Model Equation

Let C_{it} denote source commitments in millions of constant 2023 US dollars and P_{it} population. Per-person financing is $F_{it} = 10^6 C_{it} / P_{it}$. The principal dependent variable is $H_{it} = \ln(1 + F_{it})$. The addition of one permits observed zeros without deleting them. Its interpretation depends on the stated dollar-per-person unit and is not an exact percentage change in financing at low values.

The baseline equation is:

$$H_{it} = \alpha_i + \lambda_t + \beta(T_{i,t-1} / 10) + \gamma \ln(Y_{i,t-1}) + \delta D_{i,t-1} + \varepsilon_{it}.$$

Here T is openness, Y is real GDP per person, D is FDI as a percentage of GDP, α_i are country effects, and λ_t are year effects. The coefficient β compares within-country changes in lagged openness after common annual shocks and the observed controls are removed.

ISO country identifiers link the sources to a complete annual grid. Raw zeros and negative net FDI observations are retained. Country lags are created after sorting by year. The full panel contains 375 rows, including fifteen 1999 observations used only for control lags. The financing regression contains 360 rows. No interpolation, synthetic values, or manual alteration of reported finance is used.

Pooled estimation excludes country effects; the random effects model includes year indicators; the principal model includes both country and year effects. A correlated random effects comparison adds country means of the three regressors. Ordinary least squares with explicit indicators was checked against the within estimator: the main openness point estimate agrees numerically.

The model estimates an association. Country effects account for stable differences, but not changing donor priorities, reforms, interest rates, or project pipelines. Year effects account for common shocks, but not each country's own trend. Using last year's openness sets a time order; it does not make openness an independent policy experiment. Reverse causality and omitted factors are still possible.

3.3 Diagnostic Tests and Final Model Selection

The joint poolability F-test rejects common intercepts ($p < 0.0001$). The clustered Mundlak country-mean test also rejects the random effects orthogonality restriction ($p = 0.0268$). These results, together with the goal of within-country comparison, favour the country/year fixed effects specification. The conventional F-test assumes homoskedastic errors and is supplementary rather than definitive.

Breusch–Pagan does not reject heteroskedasticity, and the first-difference residual test does not reject its no-serial-correlation restriction. Neither non-rejection proves independent, homoskedastic errors. The approximate Pesaran CD test flags cross-country dependence; demeaning and short samples can affect its reference distribution.

Clustered inference uses the finite-group covariance correction and a t reference with $G - 1$ degrees of freedom (14 in the main sample). Inference compares country clustering, Driscoll–Kraay covariance (Bartlett kernel, bandwidth three), and a restricted wild country-cluster bootstrap with

9,999 Rademacher draws (seed 311). The bootstrap addresses few-cluster inference, not dependence between countries. Driscoll–Kraay also has finite-sample limits with only 24 years (Cameron et al., 2008; Driscoll & Kraay, 1998).

Within-transformed VIFs range from 1.008 to 1.019, indicating little collinearity. Individual ADF tests reject a unit root at 5% for 11/15 finance-level series, 2/15 trade series, and 15/15 finance-change series. These short-series tests do not establish joint panel stationarity. First-difference and country-trend models are consequently reported as substantive sensitivity checks.

Table 2. Diagnostic and inference checks

Diagnostic	Statistic	p-value
Poolability F (country and year effects jointly)	9.6870	<0.0001
Mundlak country-mean joint Wald (clustered)	9.1946	0.0268
Breusch-Pagan heteroskedasticity (exploratory)	1.0484	0.7895
FD residual AR(1), H0 coefficient = -0.5	1.4940	0.1574
Pesaran CD residual test (approximate)	-2.6889	0.0072
Wild country-cluster bootstrap trade H0=0	2.8787	0.0734

4. Results and Discussion: Principal Model

In the main fixed effects model, the openness coefficient is 0.1080. Its country-clustered standard error is 0.0375, the p-value is 0.0121, and the 95% confidence interval runs from 0.0275 to 0.1885. This supports a positive association under the baseline assumptions. The bootstrap and alternative models show why it should not be treated as a settled finding.

A ten percentage point increase in preceding-year openness is associated with a 0.108 increase in ln(1 + financing per person). Under the fitted transformation, one plus financing is multiplied by exp(0.108), approximately 1.114. This is an 11.4% change in one plus finance, not necessarily an 11.4% change in finance itself. If an illustrative starting amount is one dollar per person, the fitted change is about 0.228 dollars, holding other regressors and effects fixed.

The income coefficient is negative and the FDI coefficient is positive. One possible explanation is that public development finance responds to recipient needs. FDI may also reflect broader financial connections or economic conditions missing from the model. These explanations cannot be established from this regression, and the FDI measure is not specific to renewable energy.

The wild-cluster bootstrap gives p = 0.0734, which falls above the 5% threshold but below 10%. Driscoll–Kraay inference gives a smaller standard error and p < 0.0001. These methods make different assumptions, so the difference matters. With only fifteen country clusters, the evidence is less certain than the baseline p-value alone suggests.

The two-way within R^2 is 0.1074: the observed regressors explain about 10.7% of the financing variation remaining after country and year effects are removed. Financing also depends on donor priorities, project pipelines, and policy conditions that are outside the model.

Table 3. Principal fixed effects financing model

Regressor	Coefficient	Cluster SE	Cluster p	DK SE	DK p
Lagged openness / 10	0.1080	0.0375	0.0121	0.0216	<0.0001
Lagged ln income/person	-0.8754	0.3810	0.0375	0.1900	<0.0001
Lagged FDI / GDP	0.0671	0.0224	0.0097	0.0226	0.0033

4.1 Descriptive Evidence

Average clean energy commitments are 2.66 constant dollars per person, while the median is only 0.60. The maximum is 47.82 dollars per person. Average total commitments are about 284 million constant dollars per country-year. The gap between means and medians reflects lumpy finance and project timing, supporting the use of a transformation while also cautioning against a smooth annual interpretation.

Lagged openness averages 68.7%, ranging from about 21.0% to 220.4%. This spread reflects economic size and integration across the purposive sample. There are thirteen reported zero-finance observations. A zero means zero recorded commitments in this source, not zero domestic renewable investment or zero clean energy activity.

Figure 1 illustrates five financing trajectories; all fifteen countries are retained in the regression and source files. The variation is much less smooth than renewable generation. Commitments can be booked before construction and disbursement, so a finance peak need not coincide with an electricity-output increase.

Table 4. Descriptive statistics, 360 country-year observations

Variable	Mean	SD	Minimum	Maximum
Commitments (2023 US$/person)	2.6570	5.2822	0.0000	47.8158
Commitments (2023 US$ million)	284.2695	462.4547	0.0000	3309.4800
Lagged openness (%)	68.7333	41.7431	20.9822	220.4068
Real GDP/person (2015 US$)	5334.8028	3023.0321	756.7041	14932.8851
FDI inflows (% GDP)	2.6408	1.6328	-2.7574	9.6810

4.2 Visual Evidence

Figure 1 describes recorded commitments; it does not establish a trade mechanism. Figure 2 contrasts baseline and more restrictive specifications. Intervals are conventional country-clustered 95% intervals. The first-difference model changes both outcome and regressor variation, so its coefficient is a sensitivity comparison rather than an identical estimand.

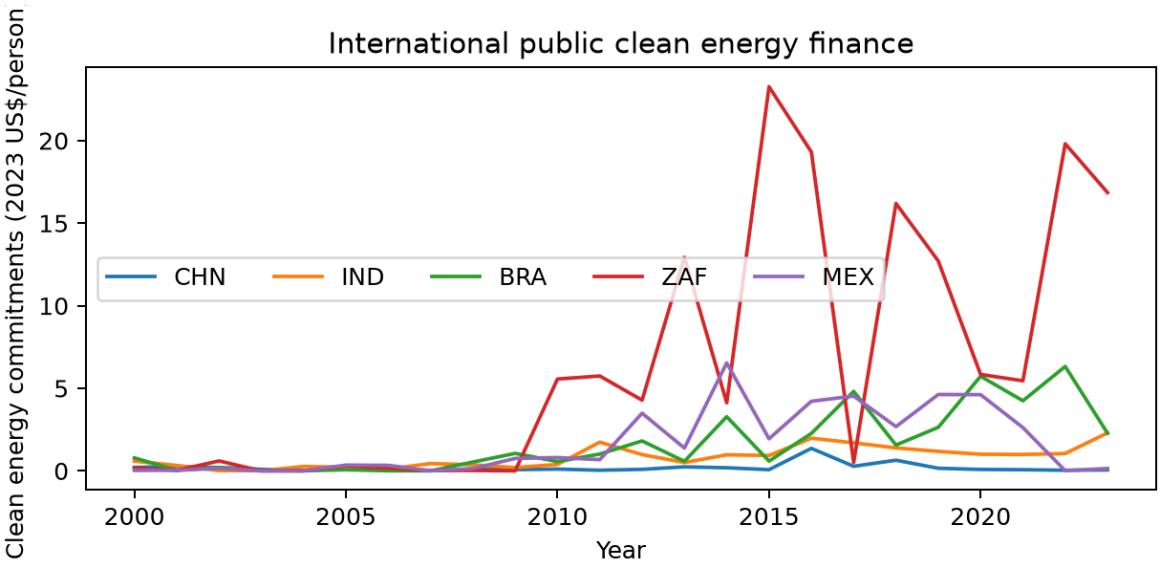


Figure 1. International public clean energy financing commitments

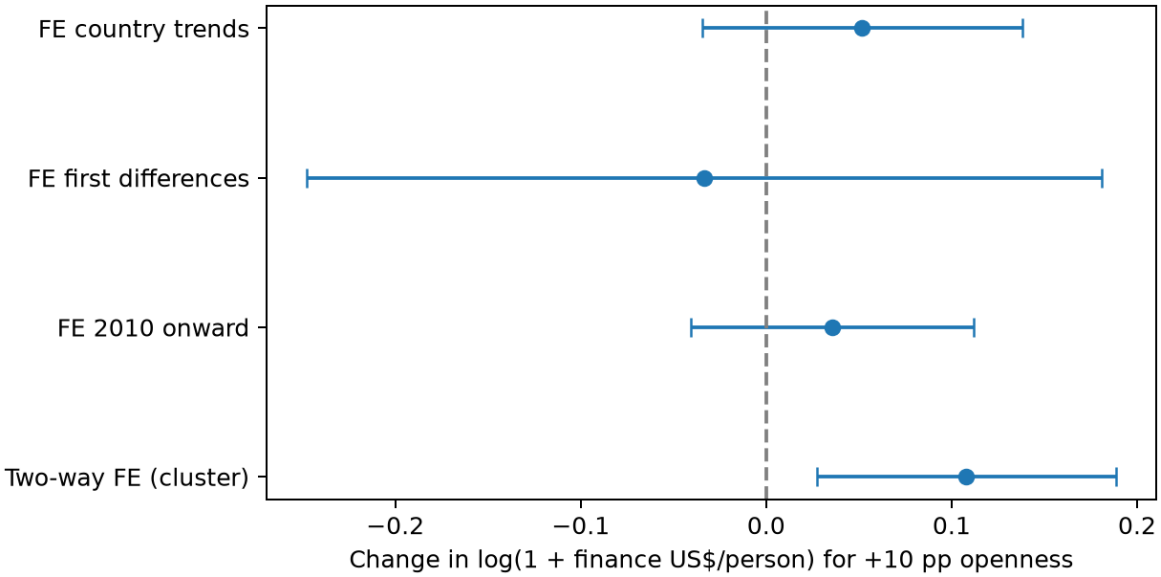


Figure 2. Openness association under alternative specifications

4.3 Robustness Results

The pooled and random effects coefficients are negative and imprecise, whereas baseline fixed effects are positive. The model diagnostics help explain why a between-country comparison is not interchangeable with the within-country relationship. Excluding China, restricting the end year to 2019, using current controls, and applying an inverse hyperbolic sine transformation preserve the positive conventional result.

Important restrictions weaken it. Starting in 2010 gives $\beta = 0.0356$ and $p = 0.3321$; country trends give $\beta = 0.0518$ and $p = 0.2183$; first differences give $\beta = -0.0334$ and $p = 0.7431$. First differences use 345 observations. Leave-one-country-out estimates remain positive, but excluding Malaysia gives $p = 0.1785$. Full coefficients and intervals are included in the research package.

These checks use variations of the same dataset, not separate replications. They show that the answer depends on which changes over time the model uses. The positive baseline estimate is worth investigating, but it is not a result that holds across all the specifications tested.

Table 5. Openness coefficients across all financing specifications

Specification	Openness β	SE	p-value
Pooled (no effects)	-0.0250	0.0264	0.3592
Random effects + years	-0.0231	0.0171	0.1981
Two-way FE (cluster)	0.1080	0.0375	0.0121
Two-way FE (DK)	0.1080	0.0216	<0.0001
FE excluding China	0.1123	0.0371	0.0096
FE through 2019	0.1370	0.0428	0.0064
FE inverse hyperbolic sine	0.1372	0.0474	0.0117
FE 2010 onward	0.0356	0.0355	0.3321
FE log total finance	0.1634	0.0759	0.0493
FE current controls	0.1131	0.0364	0.0078
FE first differences	-0.0334	0.0999	0.7431
FE country trends	0.0518	0.0402	0.2183

4.4 Economic Interpretation and Limitations

The baseline result is compatible with technology-access and financing channels: international integration may help recipients obtain equipment and connect projects with public lenders. Yet this interpretation is only one possibility. Donors can allocate funding to strategic trade partners, policy reforms can affect both openness and finance, and project commitments can themselves influence future imports. Lagged openness does not exclude these channels.

The financing measure, sample, and model each limit what can be concluded. International public finance leaves out much of total clean energy spending. Commitments do not tell us when money was paid or a plant was completed. The fifteen countries were deliberately selected, and one common coefficient cannot describe every emerging economy's response. Technology-specific financing effects are not estimated.

The country-trend and first-difference results are especially relevant because openness is persistent and most individual trade series do not reject an ADF unit-root null. Fixed effects alone do not resolve non-stationarity. These alternative specifications do not reproduce the baseline significance, so the paper avoids presenting the levels regression as a proven long-run equilibrium or using a cointegration label without a valid test.

Supplementary analysis uses solar and wind generation data from Our World in Data. With the same countries, controls, and period, the deployment-growth openness coefficient is 0.0059 with a conventional country-clustered p-value above 0.8. It neither establishes an immediate deployment response nor refutes the financing result, because the outcome and timing differ. Its complete transformations, estimates, and diagnostics are supplied separately. Generation is never relabelled as investment.

The baseline association is positive, but several checks weaken confidence in it. This does not show that liberalising trade causes more total clean energy investment. It also does not show that trade policy is ineffective. Either conclusion would require evidence beyond this study.

5. Conclusion and Policy Implications

The starting question was straightforward: does trade openness encourage clean energy investment in emerging markets? There are good reasons to expect a connection. Renewable projects often depend on imported equipment, technical knowledge, and finance from abroad. Easier trade may help developers obtain these inputs. But trade also includes fossil fuels and energy-intensive goods, so a higher trade ratio does not automatically mean more renewable investment. The results need to be read with that distinction in mind.

The report uses fifteen economies over 2000–2023, giving 360 country-year observations. Its financing measure comes from the UN SDG database, drawing on OECD and IRENA sources. It records international public commitments for clean energy in constant 2023 dollars. Dividing by population makes country sizes more comparable. World Bank data provide openness, income, FDI, and population. The renewable technology total is used once, rather than added to its components. Reported zeros are kept, missing values are not filled in, and all observations needed

for the main model are available.

The baseline model finds a positive association between the previous year's trade openness and financing per person. A ten percentage point increase in openness is associated with a 0.108 increase in the logarithm of one plus finance per person. Country-clustered inference rejects a zero coefficient. The model tests also support accounting for country differences: pooled and random effects estimates have different signs from the fixed effects result. This matters because comparing different countries is not the same as examining changes within a country over time.

That positive estimate is only part of the answer. The wild-cluster bootstrap does not reject zero at five percent. The relationship is also statistically uncertain when country trends are added, the sample begins in 2010, or first differences are used. Dropping Malaysia weakens significance too. These results make it difficult to claim that the baseline association holds across reasonable alternative specifications. The separate generation analysis provides no clear evidence of a corresponding rise in solar and wind output. Financing commitments may precede construction, and some may not translate into completed projects within the period examined.

For policy, the useful question is which part of trade could help a renewable project. Tariffs on equipment, customs delays, technical standards, and access to procurement are more specific than the overall trade-to-GDP ratio. Governments could examine these costs using project records before choosing an intervention. Public lenders should also report commitments, disbursements, construction progress, and operating output separately. That would make it easier to see whether announced finance becomes a working asset. Reliable grid connections and credible electricity purchase contracts may help, but their effects were not estimated here. These suggestions follow from the possible mechanisms and the study's limits, rather than a demonstrated policy effect.

The answer is therefore cautious. More openness is associated with higher international public clean energy finance in the baseline model, but the result is sensitive and does not establish causation. It says little directly about total domestic and private investment. A stronger study would use broader project expenditure data, distinguish financing sources and technologies, and examine clearly identified tariff reforms or external supply changes. With the evidence available here, it is reasonable to investigate the trade–finance link further. It would be premature to conclude that greater openness necessarily increases total clean energy investment.

Appendix: Individual-Series Stationarity Diagnostics

The ADF null is a unit root. A constant and linear trend are included for trade and transformed financing levels; finance changes include a constant. AIC selects up to two lags. Trade has 1999–2023 observations, finance levels 2000–2023, and finance changes 2001–2023 before lag adjustment. These are short individual-series diagnostics, without multiple-testing adjustment or a cross-sectionally robust panel reference distribution.

All countries contribute 24 observations to the principal regression. The complete ADF statistics, selected lags, and effective counts are in outputs/adf_exploratory.csv. The main-model sample audit, all tested coefficients, and raw source responses are included in the companion reproducibility package.

Table A1. All individual ADF p-values

Country	Trade ADF p	Finance ADF p	Finance-change ADF p
BRA	0.7090	<0.0001	<0.0001
CHN	0.4062	0.6334	<0.0001
IND	0.7882	0.0082	<0.0001
IDN	0.0006	0.0010	<0.0001
MEX	0.0830	0.9960	<0.0001
TUR	0.9473	0.1317	<0.0001
ZAF	0.0677	0.0045	<0.0001
THA	0.0896	0.0015	0.0004
MYS	0.7488	<0.0001	<0.0001
PHL	0.8441	0.0165	<0.0001
VNM	0.3130	0.0002	<0.0001
EGY	0.3130	0.0003	<0.0001
MAR	0.0946	0.0092	0.0002
COL	0.0184	0.0063	<0.0001
PER	0.6237	0.2956	0.0003

References

Antweiler, W., Copeland, B. R., & Taylor, M. S. (2001). Is free trade good for the environment? *American Economic Review, 91*(4), 877–908. https://doi.org/10.1257/aer.91.4.877

Appiah-Otoo, I., Ampah, J. D., & Chen, X. (2025). Governance, financial openness, and renewable energy investment in Africa. *Journal of Policy Modeling, 47*(5), 958–976. https://doi.org/10.1016/j.jpolmod.2025.05.003

Cameron, A. C., Gelbach, J. B., & Miller, D. L. (2008). Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics, 90*(3), 414–427. https://doi.org/10.1162/rest.90.3.414

Copeland, B. R., & Taylor, M. S. (2004). Trade, growth, and the environment. *Journal of Economic Literature, 42*(1), 7–71. https://doi.org/10.1257/002205104773558047

Driscoll, J. C., & Kraay, A. C. (1998). Consistent covariance matrix estimation with spatially dependent panel data. *Review of Economics and Statistics, 80*(4), 549–560. https://doi.org/10.1162/003465398557825

Ebaidalla, E. M. (2024). The impact of taxation, technological innovation and trade openness on renewable energy investment: Evidence from the top renewable energy producing countries. *Energy, 306*, Article 132539. https://doi.org/10.1016/j.energy.2024.132539

Eyraud, L., Clements, B., & Wane, A. (2013). Green investment: Trends and determinants. *Energy Policy, 60*, 852–865. https://doi.org/10.1016/j.enpol.2013.04.039

Eyraud, L., Zhang, C., Wane, A., & Clements, B. (2011). *Who's going green and why? Trends and determinants of green investment* (IMF Working Paper No. 11/296). International Monetary Fund. https://doi.org/10.5089/9781463927301.001

Our World in Data. (n.d.). *Energy data* [Data set; legacy release]. Retrieved October 12, 2026, from https://github.com/owid/energy-data

Polzin, F., Migendt, M., Täube, F. A., & von Flotow, P. (2015). Public policy influence on renewable energy investments—A panel data study across OECD countries. *Energy Policy, 80*, 98–111. https://doi.org/10.1016/j.enpol.2015.01.026

United Nations Statistics Division. (n.d.). *Global SDG indicators database: Series EG_IFF_RANDN* [Data set]. Retrieved October 12, 2026, from https://unstats.un.org/SDGAPI/v1/sdg/Series/Data?seriesCode=EG_IFF_RANDN

United Nations Statistics Division. (2026). *SDG indicator 7.a.1: Metadata*. https://unstats.un.org/sdgs/metadata/files/Metadata-07-0a-01.pdf

World Bank. (n.d.). *World Development Indicators* [Data set]. Retrieved October 12, 2026, from https://api.worldbank.org/v2/